Answer Key & Marking Scheme

Q No.	Correct Answer	Marking Points	Total Marks
1	B) 2.16 cm	• Find total length AB = AP + PB = 2.4 + 3.6 = 6.0 cm • Since PQ || BC, ΔAPQ ~ ΔABC, giving PQ/BC = AP/AB • Calculate PQ = (2.4 / 6.0) × 5.4 = 2.16 cm	1
2	B) 4.4 cm	• By Thales' Theorem (BPT), AD/DB = AE/EC => 3/1 = 3.3/EC => EC = 1.1 cm • Calculate AC = AE + EC = 3.3 + 1.1 = 4.4 cm	1
3	D) 50°	• Find ratios of sides: AB/QR = BC/PQ = CA/PR = 1/2, so ΔABC ~ ΔRQP • In ΔABC, ∠C = 180° - (75° + 55°) = 50° • Corresponding angle ∠P = ∠C = 50°	1
4	B) 4 : 7	• The ratio of perimeters of two similar triangles equals the ratio of their corresponding sides • Ratio of perimeters = 4 : 7	1
5	D) $DC^2 = CF \cdot AC$	• In ΔABC, DE || AB => CD/AC = CE/BC • In ΔBDC, EF || BD => CF/CD = CE/BC • Equating both ratios gives CD/AC = CF/CD => $DC^2 = CF \cdot AC$	1
6	C) x = 16, y = 8.	• P and R are midpoints of segments CD and DA respectively • By Midpoint Theorem, line segments parallel to DB are half the length of DB (ratio 1:2) • Thus x = 16, y = 8	1
7	D) A is false but R is true.	• Check ratio: AD/DB = 5.7/9.5 = 3/5 and AE/EC = 4.8/8 = 3/5, so DE || BC, making Assertion (A) false • Reason (R) states the Converse of Basic Proportionality Theorem, which is true	1
8	A) Both A and R are true and R is the correct explanation of A.	• In ΔABC, ∠B = 180° - (65° + 60°) = 55° using angle sum property (Reason) • Since ΔABC ~ ΔPQR, corresponding angle ∠Q = ∠B = 55° (Assertion is true and explained by Reason)	1
9	Proof shown: $AB^2 = BC \times BD$	• In ΔABD and ΔCBA, ∠DAB = ∠ACB = 90° and ∠B = ∠B (common angle) [1 mark] • ΔABD ~ ΔCBA by AA similarity criterion [0.5 mark] • AB/CB = BD/AB => $AB^2 = BC \times BD$ [0.5 mark]	2
10	DC = 8 cm (assuming AB || CD) or DC = 4 cm (if non-parallel sides are equal)	• Identify similarity between vertically opposite triangles formed by diagonals: ΔAOB ~ ΔCOD [1 mark] • Use ratio of corresponding sides: AO/OC = BO/OD = AB/DC = 1/2 [1 mark] • Calculate DC = 2 × AB = 2 × 4 = 8 cm [1 mark]	3
11	Proof shown: AF = (1/3)AC	• Construction: Draw DG || BF meeting AC at G [0.5 mark] • In ΔADG, E is midpoint of AD and EF || DG => F is midpoint of AG, hence AF = FG [1 mark] • In ΔBCF, D is midpoint of BC and DG || BF => G is midpoint of FC, hence FG = GC [1 mark] • Conclude AF = FG = GC => AC = 3AF => AF = (1/3)AC [0.5 mark]	3
12	Proof shown: PQRS is a parallelogram	• Join diagonal BD. In ΔABD, AP/AB = AS/AD = 1/3 [1 mark] • By converse of BPT, PS || BD and PS = (1/3)BD [1.5 marks] • In ΔCBD, CQ/CB = CR/CD = 1/3, so QR || BD and QR = (1/3)BD [1.5 marks] • Since PS || QR and PS = QR, one pair of opposite sides is equal and parallel, hence PQRS is a parallelogram [1 mark]	5
13	Proof shown: ∠BAC = 90°	• Given $AD^2 = BD \times DC$ => AD/CD = BD/AD [1 mark] • In ΔDBA and ΔDAC, ∠BDA = ∠ADC = 90° and BD/AD = AD/CD [1.5 marks] • ΔDBA ~ ΔDAC by SAS similarity criterion => ∠BAD = ∠ACD and ∠ABD = ∠CAD [1.5 marks] • In right ΔABD, ∠BAD + ∠ABD = 90° => ∠BAD + ∠CAD = 90° => ∠BAC = 90° (Alternatively proved via Pythagoras Theorem: $AB^2 + AC^2 = BC^2$) [1 mark]	5
14.i	BC = 56.4 m	• In right ΔCHB, cos 45° = BH/BC => BC = BH / cos 45° = $40\sqrt{2}$ • BC = 40 × 1.41 = 56.4 m	1
14.ii	AG = 38 m	• In right ΔAGF, tan 45° = GF/AG => AG = GF / tan 45° = 38 / 1 = 38 m	1

14.iii	Perimeter ≈ 281.06 m (or 245.6 m if using ∠ACH = 60°)	• Calculate base AB = AG + GD + DH + HB = 38 + 40 + 10 + 40 = 128 m, and CH = 40 m [0.5 mark] • Calculate AC: In right ΔAHC, AH = 88 m, AC = $\sqrt{(88^2 + 40^2)} = \sqrt{9344} \approx 96.66$ m (or AC = 40/cos 60° = 80 m) [1 mark] • Perimeter = AB + BC + AC = 128 + 56.4 + 96.66 = 281.06 m [0.5 mark]	2
14.iv	152.82 m	• In right ΔEHB, cos 30° = BH/EB => EB = 40 / ($\sqrt{3}$/2) = 80 / 1.73 ≈ 46.24 m [1 mark] • Calculate total length = AF + FE + EB = 53.58 + 53 + 46.24 = 152.82 m [1 mark]	2